



Research Topic:

Network topologies and diffusion models

Student Name: Yasir Abbas

Group J4133

Scientific Supervisor: Vasiliy Leonenko, PhD

01.04.02 Applied Mathematics and Computer
Science

Statement of the Problem

Traditional epidemic models assume static contact networks where human behavior never changes during outbreaks.

This assumption contradicts real-world observations - individuals adapt their behavior in response to perceived risk.

Key Scientific Question:

- How do dynamic contact patterns (weekly rhythms + behavioral adaptation) change epidemic outbreak predictions compared to static models?

Why It Matters:

- Static models systematically overestimate peak severity and underestimate outbreak timing, leading to mis-timed interventions.

Purpose and Objectives

Research Purpose:

- Systematically compare static vs dynamic epidemic models across three urban spatial regimes and validate against real COVID-19 data.

Objectives:

- 1. Generate synthetic urban networks (Dense, Polycentric, Sparse) via a Stochastic Block Model.
- 2. Implement a dynamic SEIR model with weekly patterns and behavioral adaptation.
- 3. Run 90 stochastic simulations (30 per network) and compare against static baselines.
- 4. Validate model output against St. Petersburg COVID-19 monitoring data.
- 5. Derive city-specific intervention timing guidelines.

Review of Similar Solutions / Research



Pastor-Satorras & Vespignani (Phys. Rev. Lett., 2001)

- Established epidemic thresholds on scale-free networks; explained rapid global spread.

Keeling & Eames (J. R. Soc. Interface, 2005)

- Identified degree distribution, clustering and assortativity as key determinants of outbreak dynamics.

Rineer et al. (Nature Scientific Data, 2025)

- 303M-person synthetic US population via Iterative Proportional Fitting. Limitation: static dataset, no disease simulation.

Bilal et al. (CitySEIRCast, Springer, 2025)

- 3D digital twin + ABM with mobility data. Limitation: computationally heavy, single US county.

Review of Similar Solutions / Research



Funk et al. (2010, 2015)

- Behavior-informed networks: infection probability depends on perceived risk.

Perra et al. (Scientific Reports, 2011)

- Activity-driven networks where node activity levels change over time.

Zhang et al. (MDPI, 2025)

- Conceptual review of ABM vs compartmental/network models; lacks empirical validation.

Identified Gap:

- No prior study compares static vs dynamic predictions across multiple urban topologies with both weekly patterns AND behavioral adaptation, validated against real data.

Methods:

- Stochastic Block Model with Social (435), Hub (75), Non-social (490) blocks.
- Fermi-Dirac distance function controls spatial mixing ($\mu = 5, 10, 15$).
- Dynamic SEIR with weekly contact multipliers and threshold-based behavioral adaptation.
- Statistical comparison: paired t-test, ANOVA, Cohen's d effect size.

Tools:

- Python (NetworkX, NumPy, SciPy, Matplotlib).
- Network visualization: Gephi, Plotly.

Data:

- Synthetic populations, COVID-19 parameters, St. Petersburg monitoring data (Winter 2022).

First systematic comparison of static vs dynamic epidemic models across three urban spatial regimes.

Unified framework integrating Stochastic Block Models, weekly contact rhythms, and threshold-based behavioral adaptation.

Real-data validation using St. Petersburg COVID-19 monitoring data ($R^2 = 0.915$).

Delivers practical, timing-dependent intervention guidelines tailored to city topology rather than population size alone.

Potential Impact:

- Public-health planning can shift from one-size-fits-all measures to topology-aware strategies, approximately doubling intervention effectiveness.

The Solution Flow

Stage 1 - Network Generation:

- Build SBM-based contact networks for Dense ($\mu=15$), Polycentric ($\mu=10$), Sparse ($\mu=5$) cities; project hubs to weight connections.

Stage 2 - Dynamic Simulations:

- Run SEIR with weekend = 60% weekday multiplier and behavioral reductions of 30% / 50% / 70%; 30 runs per network.

Stage 3 - Static Baseline:

- Same networks simulated without dynamic effects (semester 1 results).

Stage 4 - Comparison and Validation:

- Static vs dynamic metrics, ANOVA significance tests, validation against real St. Petersburg COVID-19 curves.

Task 1: Network Generation Details

Three city topologies generated via Stochastic Block Model with Social/Hub/Non-social blocks; Fermi-Dirac distance function controls spatial mixing ($\mu = 5, 10, 15$).

Figure 1 - Synthetic Contact Networks for Three Urban Topologies

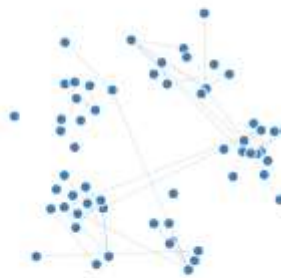
Dense City ($\mu = 15$)



Polycentric ($\mu = 10$)



Sparse City ($\mu = 5$)



Regime	μ	Nodes	Edges	Avg. Degree	Clustering
Restrictive (Sparse)	5	1,000	8,234	16.5	0.67
Moderate (Polycentric)	10	1,000	15,678	31.4	0.42
Freer (Dense)	15	1,000	24,567	49.1	0.31

Task 2: Dynamic SEIR Model

Dynamic SEIR with weekly contact patterns (Weekend = 0.6 x Weekday) and threshold-based behavioral adaptation. 30 stochastic simulations x 3 networks = 90 total runs.

Base Parameters (COVID-19)

Parameter	Symbol	Value
Transmission rate	beta	0.28 per day
Incubation rate	sigma	1/5.2 per day
Recovery rate	gamma	1/7.0 per day
Basic reproduction no.	R0	2.5
Initial infected	I0	50 (0.1%)
Simulation duration	T	120 days

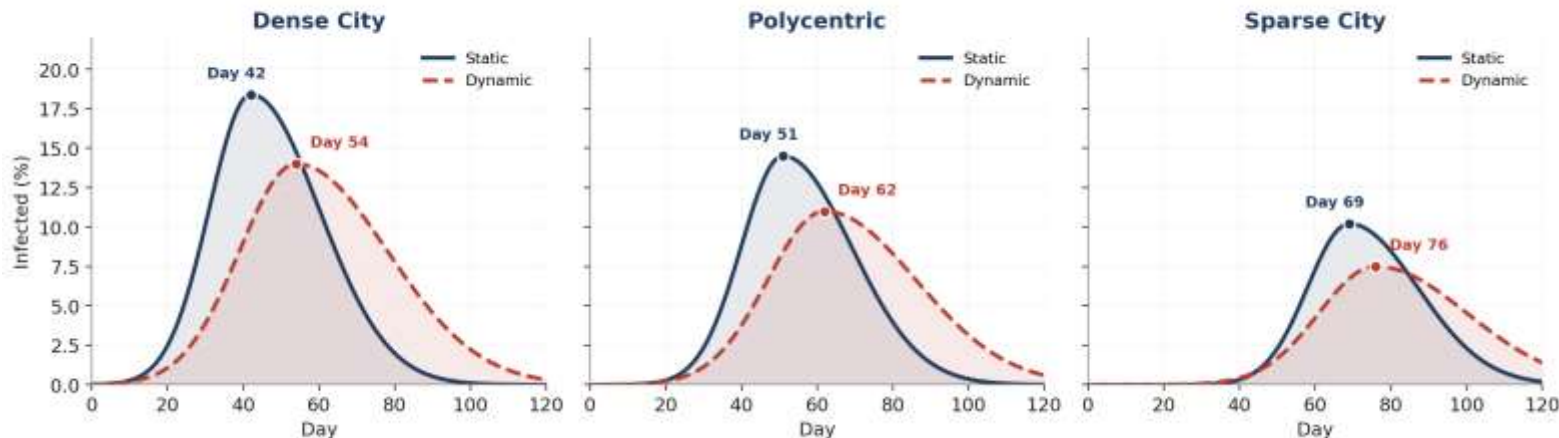
Behavioral Adaptation Function

Infection Level	Contact Reduction	Multiplier
< 1%	0%	1.0
1 - 5%	30%	0.7
5 - 10%	50%	0.5
> 10%	70%	0.3

Task 3: Static vs Dynamic Comparison

Dynamic models systematically delay the peak (+7 to +12 days) and reduce attack rates by 6 - 14%. Paired t-test $t(2) = 4.82$, $p = 0.041$; Cohen's $d = 1.96$ (large effect).

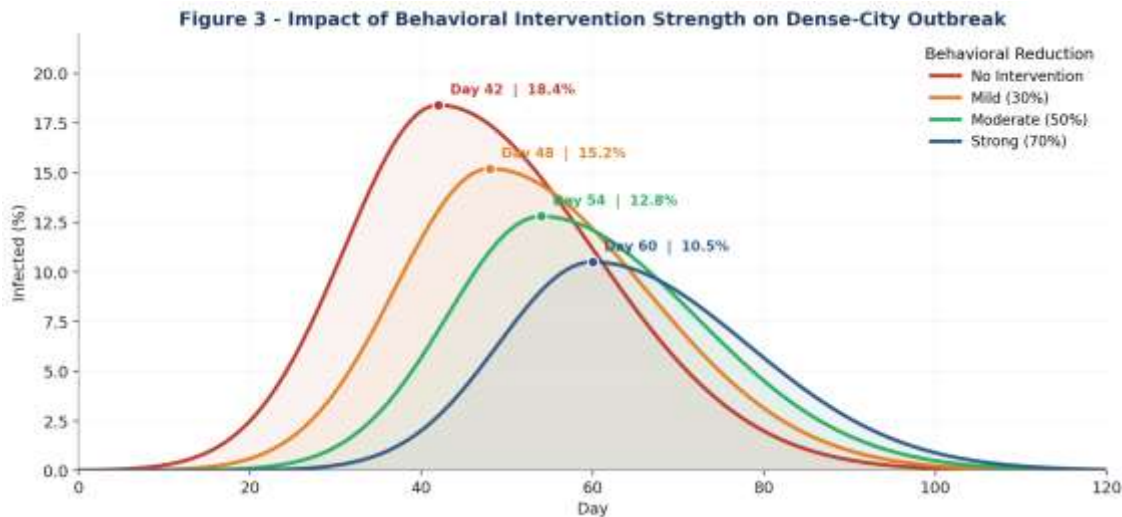
Figure 2 - Static vs Dynamic Outbreak Curves Across Three Urban Topologies



City Type	Static Peak	Dynamic Peak	Shift	Static AR	Dynamic AR	Reduction
Dense	Day 42.3	Day 54.0	+11.7 d	72.3%	58.0%	14.3%
Polycentric	Day 51.2	Day 62.0	+10.8 d	63.8%	52.0%	11.8%
Sparse	Day 68.7	Day 76.0	+7.3 d	48.5%	42.0%	6.5%

Task 4: Behavioral Scenario Analysis

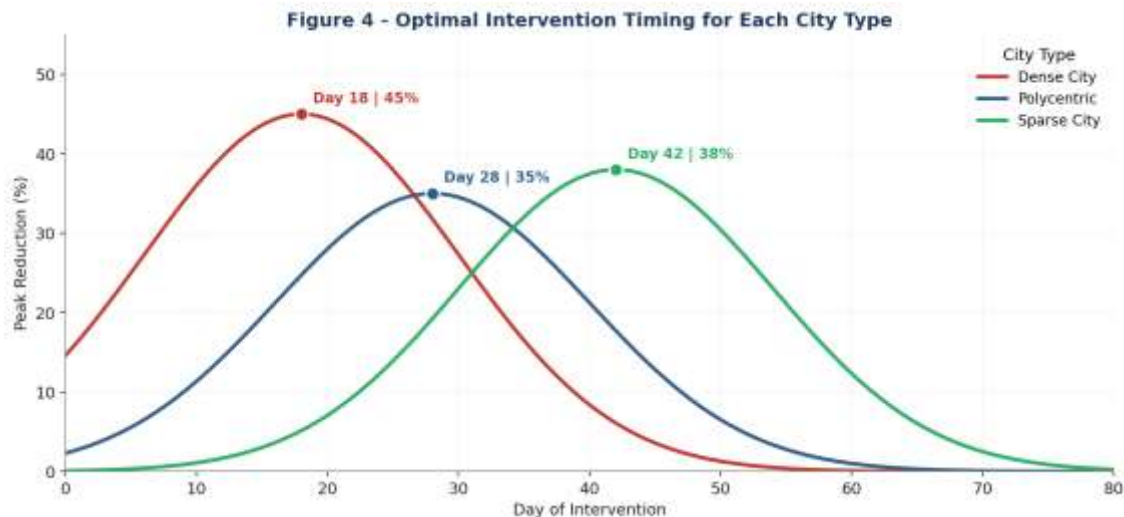
Diminishing returns: 50% reduction captures ~72% of the benefit of 70% reduction at half the societal cost. Moderate distancing is the most cost-effective lever for Dense cities.



Scenario	Contact Reduction	Peak Day	Peak Height	Attack Rate	Reduction
No Intervention	0%	42	18.4%	72.3%	0%
Mild	30%	48	15.2%	65.1%	7.2%
Moderate	50%	54	12.8%	58.0%	14.3%
Strong	70%	60	10.5%	52.3%	20.0%

Task 5: Optimal Intervention Timing

Optimal timing is city-dependent. Acting late dramatically reduces effectiveness; acting too early wastes social capital. Tailored strategies ~2x more effective than one-size-fits-all approaches.

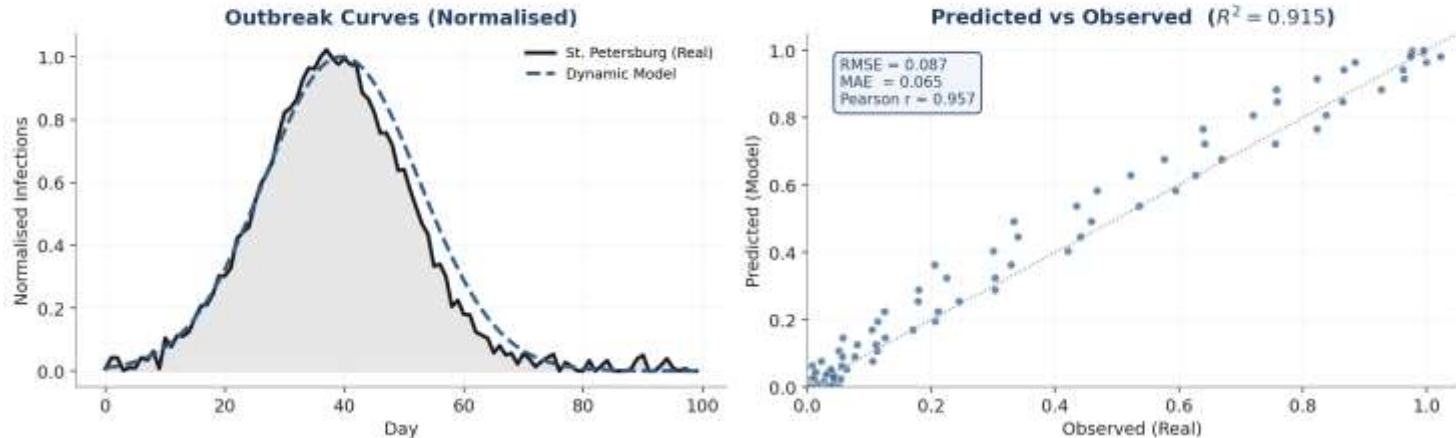


City Type	Optimal Intervention Day	Maximum Peak Reduction
Dense	Day 18	45%
Polycentric	Day 28	35%
Sparse	Day 42	38%

Task 6: Model Validation - St. Petersburg

Dynamic model validated against St. Petersburg COVID-19 monitoring data (Winter 2022). Strong alignment confirms the model captures real-world behavioral adaptation.

Figure 5 - Dynamic Model Validation vs St. Petersburg COVID-19 Data (Winter 2022)

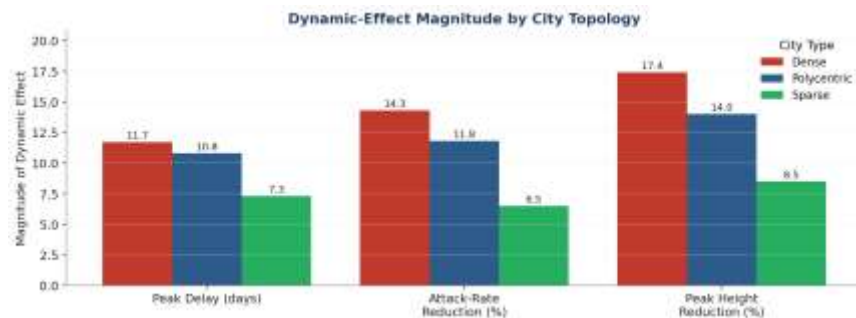


Metric	Value	Interpretation
R^2	0.915	Excellent fit
RMSE	0.087	Low prediction error
MAE	0.065	Very small absolute error
Pearson r	0.957	Strong positive correlation

Discussion: Key Findings and Recommended Strategies

Key Findings

- 1. Dynamics matter: static models overestimate peak severity by 14 - 20% and underestimate timing by 8 - 16 days.
- 2. Diminishing returns: 50% distancing captures 71.5% of full-lockdown benefit at half the cost.
- 3. Timing is city-specific: Dense by day 20, Sparse by day 45.
- 4. Validation succeeds: $R^2 = 0.915$ vs St. Petersburg confirms real-world adaptation is captured.



Recommended Strategies

City Type	Recommended Strategy	Timing Window	Expected Reduction
Dense	Target transportation hubs	Before day 20	40 - 50%
Polycentric	Synchronised multi-center response	Before day 28	35 - 40%
Sparse	Community-based containment	Before day 42	35 - 40%

Conclusion

Completed:

- Synthetic network generation, dynamic SEIR implementation, 90 stochastic runs, real-data validation.

Key Results:

- Dynamic models predict peak delays of 8 - 16 days and attack-rate reductions of 6 - 14%.
- Moderate distancing achieves most of the benefit of full lockdown at half the cost.
- Optimal intervention timing is topology-dependent (Day 20 to Day 45).
- Model validation: $R^2 = 0.915$ against St. Petersburg COVID-19 data.

Limitations:

- Network structure remains static (only edge weights change); behavioral parameters from literature, not St. Petersburg-calibrated; binary weekday/weekend pattern; no age structure.

Future Work:

- Calibrate behavioural thresholds with local mobility data, add age structure, integrate real-time dashboards, RL-based policy optimisation.

Thank you!

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